

Software Engineering Lab

UE24CS341A

Lab 1: Requirements Engineering & UML Use-Case Modelling

PROBLEM STATEMENT 6: ALUMNI MENTORSHIP & MOCK INTERVIEW PLATFORM

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1. Introduction & Scenario Review

The assigned problem is **Alumni Mentorship & Mock Interview Platform**. The platform is designed to connect students with alumni mentors working in specialized domains. It provides domain-based mentor matching, availability calendar booking, and structured post-interview scorecard recording.

The main stakeholders identified in the scenario are **Student Mentee** and **Alumni Mentor**. The Student Mentee uses the platform to find suitable mentors, view their availability, and schedule mentorship or mock-interview sessions. The Alumni Mentor provides their areas of expertise and availability, conducts mentoring or mock-interview sessions, and provides feedback through a structured scorecard.

The system should ensure that students are matched with relevant mentors, available time slots are accurately maintained, sessions can be booked without scheduling conflicts, and interview feedback is recorded securely.

2. Elicit & Draft Requirements

Brainstorm

Based on the given scenario, the following possible system functions and constraints were identified.

Possible Functions:

1. The system shall recommend suitable alumni mentors based on the student's selected domain of interest.
2. The system shall allow alumni mentors to create and update their profiles with their areas of expertise and mentoring preferences.
3. The system shall allow alumni mentors to create, update, and remove their available mentoring time slots.
4. The system shall allow students to book mentorship or mock-interview sessions with selected alumni mentors.
5. The system shall allow students to select an interview domain or track such as DSA, System Design, or HR.
6. The system shall allow alumni mentors to submit a structured post-interview scorecard containing ratings and comments.
7. The system shall allow alumni mentors to publish interview experience posts containing preparation methods and relevant information.

Possible Constraints:

1. A mentor's time slot shall not be available for booking after it has already been booked.
2. The system shall prevent double-booking of the same mentor and time slot.
3. Mentor recommendations should be based on the student's selected domain of interest.
4. The selected interview domain or track should be recorded with the mock-interview booking.
5. The mentor should be able to view the selected interview track before the session.
6. Feedback scorecards should only be submitted after the mock-interview session is completed.
7. User profile data and feedback scorecards shall be protected from unauthorized access.
8. Alumni contact information shared through interview experience posts shall only be visible according to the mentor's selected privacy settings.
9. Requested user data should support soft-deletion within 24 hours.
10. Mentor recommendations and available session slots should be displayed within a reasonable response time under normal operating conditions.

Requirements Table :-

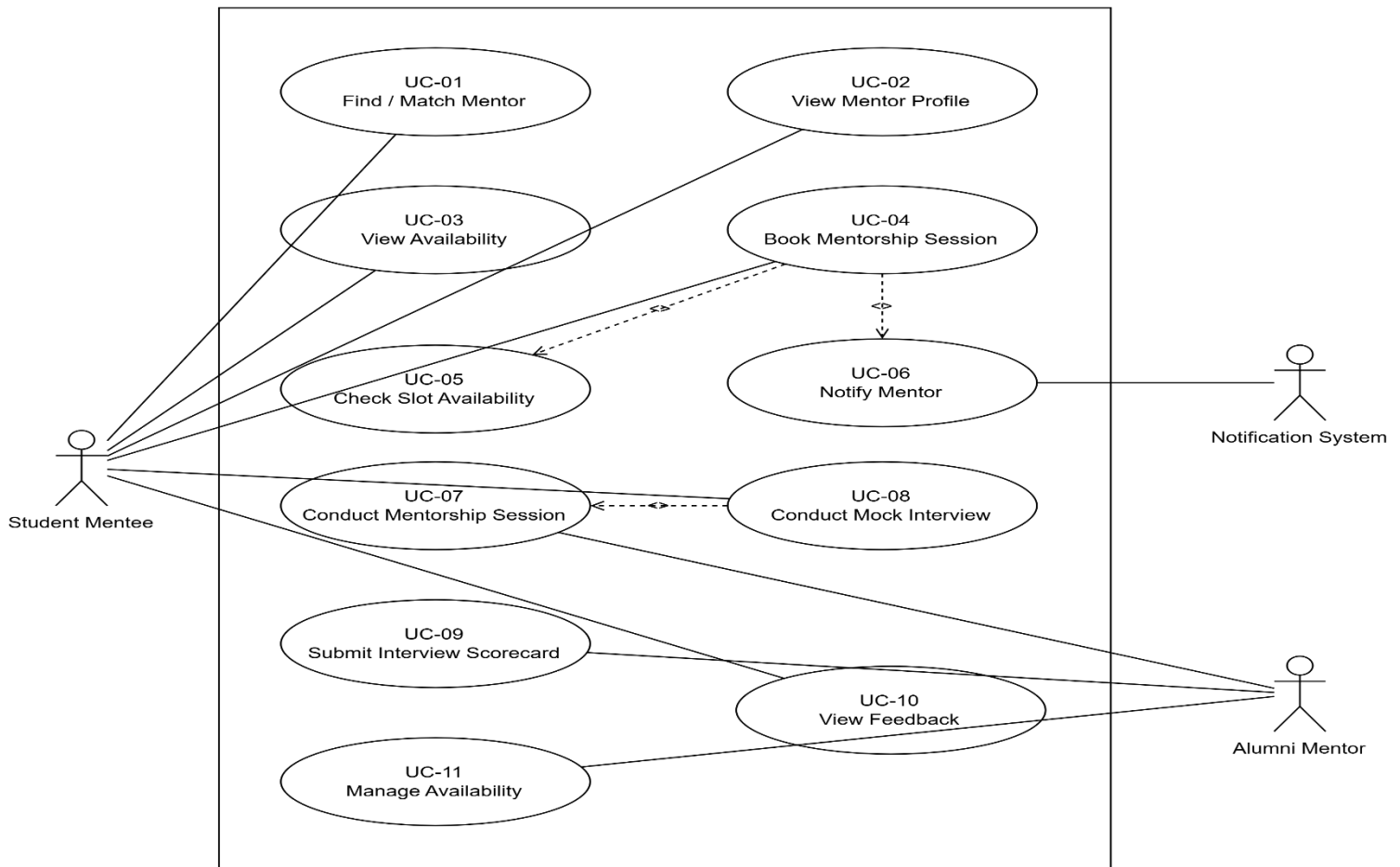
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall recommend suitable alumni mentors to students based on their selected domain of interest and allow the student to book a recommended mentor.	High	Pass: The system displays mentors whose expertise matches the student's selected domain and allows the student to select a mentor for booking. Fail: Unrelated mentors are recommended or the student cannot select a recommended mentor.	Helps students find relevant mentors efficiently through domain-based matching.
FR-002	Functional	The system shall allow alumni mentors to create, update, and manage their weekly availability slots on a shared calendar.	High	Pass: A mentor can add, modify, or remove an availability slot, and the updated slot is reflected in the student's view. Fail: Changes are not saved or already-booked slots are displayed as available.	Prevents scheduling conflicts and provides accurate booking options.
FR-003	Functional	The system shall allow students to schedule mock-interview sessions and select an interview domain or track such as DSA, System Design, or HR.	High	Pass: The selected interview track is recorded with the booking and is available to the mentor before the session. Fail: The booking is created without a track or the mentor cannot view the selected track.	Helps mentors prepare relevant questions and guidance for the selected interview track.
FR-004	Functional	The system shall allow alumni mentors to submit a structured post-interview scorecard containing ratings and comments after completing a mock interview.	High	Pass: The mentor can enter the required ratings and comments, and the submitted scorecard becomes available to the student. Fail: An incomplete scorecard is submitted or the submitted scorecard cannot be viewed by the student.	Provides students with structured and actionable feedback on their interview performance.
FR-005	Functional	The system shall allow alumni mentors to publish an "Interview Experience" post containing their company, role, preparation methods, and optional contact information for students to view.	Medium	Pass: The post is saved and displayed on the mentor's profile, and contact information is shown only when the mentor enables its visibility. Fail: The post is not displayed or contact information is shown when visibility is disabled.	Gives students access to real interview experiences, preparation strategies, and optional mentor contact.
NFR-001	Nonfunctional	The system shall protect user profile data and feedback scorecards according to privacy guidelines and support soft-deletion within 24 hours of a deletion request.	High	Pass: Requested data is soft-deleted within 24 hours and cannot be accessed through normal user interfaces. Fail: The data remains normally accessible after 24 hours.	Protects sensitive student and alumni information and supports privacy requirements.
NFR-002	Nonfunctional	The system shall restrict alumni contact information shared through "Interview Experience" posts to authenticated students and prevent it from being indexed by external search engines.	High	Pass: Unauthenticated users cannot access the contact information and the information does not appear in external search results. Fail: Contact information is accessible without authentication or appears in search-engine results.	Prevents misuse of alumni personal contact information.

3. Peer Critique & Revision

Req ID	Clarity	Testability	Relevance	Comment
FR-001	Clear	Pass/Fail can be tested	Relevant	Clearly specifies domain-based mentor matching.
FR-002	Clear	Can test slot creation/update/removal	Relevant	Availability changes should be reflected correctly.
FR-003	Clear	Can verify track is recorded	Relevant	Clearly connects mock interview with selected track.
FR-004	Clear	Can verify scorecard submission	Relevant	Specifies structured feedback after interview.
FR-005	Clear	Can verify post visibility/privacy	Relevant	Clearly defines interview experience sharing.
NFR-001	Clear	Can test deletion within 24 hours	Relevant	Addresses privacy of user and scorecard data.
NFR-002	Clear	Can test authentication/search indexing	Relevant	Protects alumni contact information.

4. UML Use-Case Diagram :-

Alumni Mentorship & Mock Interview Platform



5. Use-Case Flow:-

Use Case ID / Name	Considered UC-09: Submit Interview Scorecard
Primary Actor	Alumni Mentor
Secondary Actors	Student Mentee (recipient of feedback)
Related Requirements	FR-004 (High), NFR-001 (Privacy / soft-deletion)
Trigger	Mock interview or mentorship session (UC-07 / UC-08) has concluded

Preconditions:

1. The Alumni Mentor is authenticated and is the assigned mentor for the completed session.
2. The linked mentorship or mock-interview session (UC-06/UC-07) has a status of "Completed."
3. No scorecard has already been submitted for this session (one scorecard per session).

Postconditions:

1. A structured scorecard record (ratings + comments) is persisted and linked to the session.
2. The scorecard becomes visible to the student mentee on their feedback view (UC-10: View Feedback).
3. The session status is updated to "Feedback Submitted" and can no longer accept a second scorecard.

Main Success Scenario:

1. Alumni Mentor opens the completed session from their dashboard and selects "Submit Scorecard."
2. System presents the structured scorecard form: per-criterion ratings (e.g., Technical Depth, Communication, Problem-Solving, Confidence) on a fixed scale, plus a free-text comments field.
3. Mentor enters a rating for each required criterion.
4. Mentor enters written comments and, optionally, suggested next steps for the student.
5. Mentor selects "Submit."
6. System validates that every required rating field is filled and comments are within the allowed length.
7. System timestamps and persists the scorecard, linking it to the session, mentor, and student.
8. System updates the session status to "Feedback Submitted" and marks the record as visible to the student.
9. System displays a submission confirmation to the mentor.
10. Use case ends successfully.

Alternate Flows:

- **6a. Incomplete Scorecard**
 - 6a1. System detects one or more required rating fields are empty.
 - 6a2. System highlights the missing fields and displays "Please complete all ratings before submitting."
 - 6a3. Mentor fills in the missing fields and resumes at step 5.
- **6b. Duplicate Submission Attempt**
 - 6b1. System detects a scorecard already exists for this session (e.g., mentor resubmits from a stale page).
 - 6b2. System blocks the write and displays "A scorecard has already been submitted for this session."
 - 6b3. System redirects the mentor to the read-only view of the existing scorecard.
- **9a. Mentor Abandons Submission**
 - 9a1. Mentor navigates away before submitting.
 - 9a2. System auto-saves entered ratings/comments as a draft, retrievable next time the mentor opens the session.
 - 9a3. Session status remains "Pending Feedback" until a scorecard is successfully submitted.